Healthcare's AI Trap

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Abstract

In clinical care the routine cases do double duty: under a flat fee they are the profitable cases that cross-subsidize complex care, and they are the cases on which junior clinicians become senior ones. AI automates them first, severing both supports at once. On the financing side, an AI-native entrant that serves only the routine "cream" raises the average cost of the patients left behind above the flat fee and the pool unravels (Einav and Finkelstein, 2011; Faulhaber, 1975), before AI can treat the complex patients it abandons. We solve the skimming game from primitives: the over-skimming wedge carries a business-stealing term that mere fewness cannot remove, so efficiency is restored by *vertical integration* of the cream with the tail, with risk adjustment (Glazer and McGuire, 2000) as the price-side cure. On the workforce side, automating the routine cases starves the pipeline that produces specialists; because expertise takes a decade to build and the option of keeping it alive is never priced, the loss can be irreversible (Arrow and Fisher, 1974), and supervision and training compete for the same seniors, giving a tipping point between a healthy steady state and a low-capacity trap. The two externalities share the same cases but are not the same problem: integrating the cream with the tail removes the financing externality and trains enough seniors to replace a provider's own retirements, yet no market structure prices the option value of training above replacement, which spills across firms, so transfers and administered prices alone do not address both margins and a separate training instrument is needed. What is specific to medicine is that the cases AI takes first are at once the money that funds hard care and the making of the people who provide it.

Keywords. Artificial intelligence; automation; adverse selection; cream-skimming; risk adjustment; clinical training; ironies of automation; human capital.
JEL. I11, I18, J24, D62, L13.

1 Introduction

In clinical care the routine cases do double duty. Under a flat fee they are the profitable cases, the ones whose easy margin quietly pays for the complex care that loses money, and they are also the cases a junior clinician works through, by the thousand, to become a senior one. AI automates the routine cases first, so it pulls on both supports at once: the money that funds hard care, and the making of the people who provide it. Two supports resting on the same cases, cut by the same tool, is the whole paper.

This is not the old fear that machines replace workers (Acemoglu and Restrepo, 2018; Autor, 2015); that worry is real but not ours. Our question is narrower and specific to medicine: *when AI does the easy clinical work, who pays for the hard work, and who is left able to do it?* The headline is that two failures grow from one root. Both come from flat pricing, paying about the same for a simple visit and a hard one. The first is a money failure and the second is a people failure; they are not one problem wearing two labels, and the simple levers, pure transfers and flat administered repricing, do not reach both margins, so the policy response combines a pricing fix with a separate training instrument (Proposition 6).

The money failure comes first, and its timing is the surprise. A clinic runs on a quiet cross-subsidy: the easy patients are profitable and pay for the hard ones, who lose money. An AI-native entrant can serve the easy patients for almost nothing and simply decline the hard ones. Once it does, the patients left behind cost more, on average, than the flat fee pays, and the old practice goes broke, the provider-side version of the adverse-selection death spiral (Einav and Finkelstein, 2011; Cutler and Zeckhauser, 2000). This is one of the best-documented failures in health economics, the reason Medicare Advantage plans attract healthier enrollees and the reason risk adjustment exists at all; what is new is that AI brings it to primary care, cheaply and at scale. The surprise is when: the clinic fails well before AI can treat even one of the hard cases it walked away from (Proposition 1). It fails not because the machine took the hard work but because skimming the easy work pushed the average cost of who is left above the fee.

The people failure is slower and harder to undo. Forty years ago Bainbridge (1983) noticed that taking routine work away from a human operator also takes away the practice that keeps them ready for the rare case the machine cannot handle; her "final irony" was that the better the automation, the more one must spend to keep human skill alive. Aviation learned it the hard way, with pilots raised on autopilot who could not recover a basic stall (Air France 447). Medicine is where it bites hardest, because the cases machines cannot handle are not a rare jet upset but a large and permanent share of care, and a specialist takes about a decade to grow. The cases that grow one are exactly the cases competition automates first. We frame this as a timing mismatch, not a prophecy of doom: AI improves in months while senior capacity rebuilds over a decade, so a shortfall, once it opens, cannot be closed on the horizon that matters. It is a medium-run trap, binding for as long as the labor market's own response, the scarcity pay that would eventually pull people back into training, is slower than retirement.

The mechanisms here are classical and we claim no new one. Selection, cream-skimming, and risk adjustment are textbook (Einav and Finkelstein, 2011; Glazer and McGuire, 2000), and automation's erosion of training pipelines is an active literature of its own (Zhang, 2026; Afrouzi et al., 2026). The contribution is the coupling itself: a single set of routine cases that both funds hard care and trains the people who provide it, so one act of automation fires a financing externality and a workforce externality at once, and the two-instrument policy reads directly off that one object. The mechanisms are old; the coupled object is new. The closest sibling is the AI layoff trap of Hemenway Falk and Tsoukalas (2026), where one large firm internalizes the externality so that fewness fixes it. Here fewness is not enough: a lone entrant with no market power still steals business by pulling the cream, so the cure is not concentration but integration of the cream with the tail. That distinction also fixes what we can and cannot claim about market structure. Integrating the cream with the tail removes the money failure completely, and an integrated provider trains enough seniors to cover its own retirements (Kaiser runs residencies). What integration cannot do is supply the option value of training above replacement, because in the bad state that value spills out across firms and is a public good even for a single integrated system. So integration is necessary and sufficient for the money failure but only necessary, not sufficient, for the people failure (Proposition 6); that gap is why two instruments are needed, and why "integrate the cream with the tail" must not be misread as "concentration is good for training."

Section 2 sets up the cost-curve model and the planner's benchmark. Section 3 solves the financing externality. Section 4 solves the workforce externality. Section 5 joins the two and reads off the two-instrument policy, with a short calibration. Section 6 discusses scope and limits. Proofs are in Appendix A.

Related literature

Selection and its correction: Akerlof (1970); Rothschild and Stiglitz (1976); Einav and Finkelstein (2011) and Einav, Finkelstein and Cullen (2010) on the cost-curve framework; Cutler and Zeckhauser (2000) and Newhouse (1996) on health-market selection; Glazer and McGuire (2000) and van de Ven and Ellis (2000) on risk adjustment; Brown et al. (2014) on Medicare Advantage risk selection; Faulhaber (1975) on cross-subsidy sustainability. *Automation and human capital:* Autor, Levy and Murnane (2003), Autor (2015), Acemoglu and Restrepo (2018); Bainbridge (1983); and the human-capital-trap results of Zhang (2026) and Afrouzi et al. (2026). *AI in health care:* Dranove and Garthwaite (2022); Sahni et al. (2023). We also use the option-value-of-irreversibility theory of Arrow and Fisher (1974), Henry (1974), and Dixit and Pindyck (1994), and Arrow (1963) on the physician agency relationship. *Entry and cross-subsidy:* the financing half is the "universal service" problem, one price for everyone, the profitable customers funding the unprofitable ones, until a rival grabs the profitable ones (Armstrong 2001; Choné, Flochel and Perrot 2002; Valletti, Hoernig and Barros 2002; Panzar and Willig 1977 on why this cannot be stopped under free entry); our skim tax is their access charge, and Kong, Layton and Shepard (2024) and Shepard (2022) show the skim survives risk adjustment. *Optimal risk adjustment and its limits:* Glazer and McGuire (2000); Geruso and Layton (2020) on upcoding, which caps how finely complexity can be priced. *Over-automation externalities:* the closest siblings are Hemenway Falk and Tsoukalas (2026) and Bayraktar (2026), and the human-capital trap of Afrouzi et al. (2026); measured deskilling under AI exposure appears in Budzyń et al. (2025). The case continuum is Acemoglu–Restrepo with administered prices; the contribution relative to all is the dual-role coupling, one set of cases that funds hard care and trains its providers, so automating it fires both externalities and the two-instrument policy follows.

2 Model

Line up every patient a clinic might see, from the easiest to the hardest, and call a patient's place in that line its *complexity* c, running from 0 (a quick routine visit) to 1 (the most tangled case). The share of patients at each complexity is $f(c)$, and we count the whole population as one unit, so these shares add to 1 and the clinic's revenue, the same fee from every patient, is just r. Three facts about money, in plain terms:

- **What the clinic is paid.** Insurance pays a fixed fee r per visit, the same whether the patient is simple or complicated (the capitation / compressed-fee regime).
- **What a human costs.** A doctor handles complexity c at cost $w(c)$, which rises with how hard the case is. Lined up by complexity, $w(\cdot)$ is the clinic's cost curve.
- **What AI costs.** AI handles case c at cost $k(c)$, far below a human on the easy cases ($k(0) < w(0)$) but rising as cases get harder. On hard cases AI may still be cheaper, or it may end up costing *more* than a human would. That second case matters: when $w(c) < k(c) < r$, automating the case still earns money (the AI cost beats the fee) but wastes resources (a human would have been cheaper). And AI is only safe up to a frontier θ, which we hold fixed in Section 3 and let rise in Section 4.

The one assumption that drives everything is that the flat fee sits between the cost of the easiest and the hardest case:

Assumption 1. *$w(0) < r < w(1)$, with the* break-even *case $c^\dagger$ defined by $w(c^\dagger) = r$; and $r > \bar{w} \equiv \int_0^1 w(c)f(c)\,dc$.*

In words: easy cases (below $c^\dagger$) make money, the *cream*; hard cases (above it) lose money; and a clinic holding the whole panel still breaks even, because the fee beats the *average* cost, $r > \bar{w}$ (a bar over a quantity denotes its average over all patients). So the easy cases quietly pay for the hard ones, the cross-subsidy from the introduction. The obvious objection is that this is just a pricing mistake: if insurance paid more for sicker patients, the fee would track each patient's expected cost, the easy cases would stop being unusually profitable, and there would be no cream to skim. That fix has a name, *risk adjustment*, and it is the right idea. The catch is that it cannot be carried down to the individual patient. A payer cannot observe any one patient's true cost; it can only approximate it from coarse, gameable signals such as diagnosis codes, so payment rises in broad brackets instead of tracking cost case by case. Read the flat fee r, then, as the payment *within* one such bracket; Assumption 1 is about the margin still falling with complexity *inside* a bracket and turning negative on the most complex panels, as in complex Medicaid primary care. So risk adjustment shrinks the cream but never removes it: it captures only a fraction (the $\phi < 1$ of Section 5) of the true cost difference, the unpriced remainder is still cream, and selection is found to persist wherever risk adjustment is applied (Brown et al., 2014). The brackets do get repriced over time, but slowly and under budget caps; the mechanism needs only that they lag the AI frontier. The flatness is therefore structural, not a correctable error, which is why the externality survives the obvious fix. This is the cost-curve picture of Einav and Finkelstein (2011) seen from the provider side: a flat price line r cutting an upward-sloping cost curve $w(c)$.

We will call r an *administered* fee, meaning one set by formula rather than negotiated. The cleanest examples are Medicare and Medicaid, and in our calibration r is specifically the professional payment for an office visit under the Medicare physician fee schedule, not a hospital facility rate. But the payer is not what does the economic work. The mechanism needs only one thing: the fee does not move with the individual patient's cost c. That holds well beyond government pricing. Medicare visit codes are flat within each coarse tier, commercial fee-for-service is flat per code, and capitation, a single payment per member, is flattest of all. So the cross-subsidy does not require prices to be government-set, only that they not track case-level cost; the administered case is simply the cleanest and most defensible example.

The AI channel carries a small convenience edge ε, it is faster and easier to reach, while a patient's willingness to pay for a human rises with complexity (higher stakes, more need for judgment and reassurance). So on the easy cases patients self-select into the AI channel, and the hard cases stay with human care. This one sentence is all we ask of demand, and it is what lets an AI-native entrant skim the routine band by patient choice alone, with no screening: the share it captures grows with ε, and we study the limit in which the whole feasible routine band migrates.

The routine (low-c) cases do two jobs at once, which is the whole paper: **(i) they pay the bills**, carrying the margin $r - w(c) > 0$ (or $r - k(c)$ once automated) that funds the hard cases; and **(ii) they train the doctors**, being the cases juniors learn on to become seniors (made precise in Section 4). Automating them hits both jobs together.

It helps to keep **three frontiers** straight: θ, what AI *can* do; a, what the market *chooses* to automate; and θ_{eff}, what the supply of senior doctors *permits*. The gap between the first two is the money problem (Section 3); the gap between θ and θ_{eff} is the trap (Section 4).

Let treating a case be worth $v(c)$ to society, more than it costs. As a benchmark, ask what an ideal *planner* would do, an imagined public-minded decision-maker who could assign every case as it liked and would do so to make the whole system as efficient as possible, ignoring who profits. This is not a policy proposal or a real authority; it is simply the yardstick for the best the system could do, and the gap between it and the market is what we will call the inefficiency. Such a planner would hand the easy cases to AI and keep humans on the rest, for a total resource cost of

$$C^*(\theta) = \int_0^\theta k(c)\, f(c)\, dc + \int_\theta^1 w(c)\, f(c)\, dc. \tag{1}$$

Read the two terms as two bills: what AI costs on the easy cases it takes, plus what humans cost on the rest. Automating helps only where AI is actually cheaper: where $k(c) < w(c)$ it lowers cost, and where $k(c) > w(c)$ it does not. So if AI is cheaper than humans across the whole feasible band the planner automates all of $[0, \theta]$; otherwise it stops automating where k rises past w. This test, automate a case only where the machine is the cheaper way to do it, is the efficiency benchmark the rest of the paper measures against.

Observation 1. *A single provider that does exactly this, runs AI wherever it is cheaper than a human, keeps humans on the rest, still turns a profit and pays for the hard cases out of those savings:*

$$\Pi_{\text{int}}(\theta) = (r - \bar{w}) + \int_0^\theta \big(w(c) - k(c)\big)^+ f(c)\, dc \;>\; 0. \tag{2}$$

So any trouble that follows is about who *automates, not about automation itself.*

3 The financing externality

The force in this section is not one we invent; it is one of the best-documented failures in health economics. Physician-owned specialty hospitals take the profitable cardiac and orthopedic cases and leave the safety-net hospital the complicated ones (Barro, Huckman and Kessler, 2006). Medicare Advantage plans end up with healthier enrollees even after risk adjustment (Brown et al., 2014; Cabral, Geruso and Mahoney, 2018). The entire apparatus of risk adjustment exists precisely because everyone knows skimming happens if you let it. We are not inventing the force; what is new is its reach. AI makes skimming cheap and scalable in primary care, a tier that used to be too low-margin and too bundled to skim profitably. And the entrant does not refuse anyone: selection almost never looks like turning a patient away, it looks like which conditions you market to, which plans you take, and what your intake can actually handle. The direct-to-consumer telehealth platforms now built around a single condition or two do exactly this, politely, through what they market and what their intake handles rather than by refusal.

Now let entrants in. An AI-native entrant earns $r - k > 0$ on every easy case, so it takes the easy band $[0, \theta]$ and leaves the incumbent the hard tail $(\theta, 1]$. The incumbent cannot just keep the cream to fund the tail, because an entrant living on the cream alone is viable, and under free entry that alone is enough to break the cross-subsidy (Faulhaber, 1975). How does the entrant win the easy cases, when the fee is the same for everyone? Not on price but on convenience: the small edge ε of Section 2 is enough for the easy patients to choose it, even if the incumbent also adopts AI. So the entrant does not even have to *measure* complexity to skim. The patients sort themselves, and whatever extra sorting the entrant does runs through product design, intake screens, and referral-out, not through turning anyone away.

This points to an asymmetry we lean on later. The entrant can select with a rough, self-enforcing filter, but the payer, to pay more for complex cases, needs a complexity measure it can *audit*, and it does not fully have one. That gap is what leaves risk adjustment imperfect ($\phi < 1$ in Section 5; the ceiling on it is upcoding, Geruso and Layton 2020). We take the entrant to capture the whole routine band, the share $s = F(\min\{\hat{c}, \theta\})$ at its largest, where $\hat{c}$ is the highest-complexity case patients still self-select into the AI channel on the convenience edge ε, capped by the AI capability frontier θ (the entrant can take a case only if patients choose it *and* AI can safely handle it). If it captures less, the residual pool keeps some cream and the collapse point $\bar{\theta}$ only arrives later, so full migration is the conservative case. The mechanism also needs the fee r to hold still while the residual pool's costs rise, which it does, since fee updates are slow and budget-capped. What the incumbent is left earning on the tail is

$$\Pi_I(\theta) = \int_\theta^1 \big(r - w(c)\big) f(c)\, dc, \qquad \Pi_I'(\theta) = -\big(r - w(\theta)\big) f(\theta). \tag{3}$$

That leftover panel survives only while its average cost stays under the fee, $\mathbb{E}[w(c) \mid c > \theta] \leq r$. Call the point where it stops surviving the **collapse point** $\bar{\theta}$, where the leftover profit hits zero, $\Pi_I(\bar{\theta}) = 0$ (equivalently, the average cost of the cases above $\bar{\theta}$ just equals r).

Proposition 1 (Early unraveling)**.** *The leftover panel goes broke* before *AI can even treat the hard cases it abandoned: $\bar{\theta} < c^\dagger$. For $\theta > \bar{\theta}$ the leftover cases cost more, on average, than the fee pays, and the pool cannot survive on flat payment, the provider-side version of the adverse-selection death spiral.*

It is worth pausing on what $\bar{\theta}$ means. Put difficulty on a scale from 0, the simplest visit, to 1, the most complex case, with break-even at $c^\dagger = 0.5$, the difficulty where a case stops paying for itself. The collapse point sits well below that, around 0.32 in our illustrative numbers (Figure 2; illustrative, not an empirical estimate): squarely in routine-care territory. So the clinic does not fail because AI got good at hard medicine. It fails while AI is still only doing routine work, because a competitor took the easy patients who were paying the bills. That timing is the robust part, not an artifact of the example: Proposition 1 puts the break before break-even for *any* margin. How many *patients* that represents does depend on the margin. Here, with a comfortable margin, about the easiest half of visits sit below $\bar{\theta}$; but real primary care runs on low-single-digit margins, where the patients left behind can absorb far less skimming, and collapse comes after only a small slice of routine volume is gone, well under a fifth (Appendix B).

Corollary 1 (Welfare is flat, then a widening leak)**.** *For $\theta \leq \bar{\theta}$ (full-band automation, $a = \theta$) the allocation equals the first best and skimming is purely redistributive: the same patients get the same care at the same total cost, and only who pockets the profit changes, so society loses nothing. For $\theta > \bar{\theta}$ the residual pool runs a deficit $-\Pi_I(\theta) = \int_\theta^1 (w - r) f\, dc > 0$ and society loses*

$$\min\Big\{ \textstyle\int_\theta^1 (v - w) f\, dc, \quad \delta\,(-\Pi_I(\theta)) \Big\}, \tag{4}$$

either the lost value of the complex care abandoned, or the deadweight cost of bailing the pool out with public money, which costs $1 + \delta$ per dollar raised. Since $-\Pi_I(\bar{\theta}) = 0$ by the definition of $\bar{\theta}$, this loss is zero at $\bar{\theta}$ and opens linearly *from there, at rate $\delta\,(r - w(\bar{\theta}))\, f(\bar{\theta}) > 0$: welfare stays continuous in level but its slope drops at $\bar{\theta}$, a kink, not a cliff. The hole opens at first order, not as a vanishing triangle, because the last cream case skimmed at $\bar{\theta}$ still earns a positive margin, $w(\bar{\theta}) < r$. This is the same fact as the timing result $\bar{\theta} < c^\dagger$ (Proposition 1): collapse arrives while the marginal case is still profitable, so pulling it tears a first-order hole. A true cliff returns only if you add a friction: an incumbent exiting and destroying organizational capital, or a rescue that arrives only after a lag during which the tail goes unserved. That is the financing version of the human-capital irreversibility in Section 4. (The* min *just says society takes the cheaper of abandoning the tail or rescuing it. And the standing option of rescue is itself a mild incentive to skim a little more, a small effect we do not price.)*

It is fair to ask why this matters if no clinic ever literally files for bankruptcy. A model is not a forecast; it is a way of finding which thing is load-bearing. The result worth saying out loud is the timing: the danger is not that AI gets good enough to replace the hard work, it is that selective adoption can break the funding model while AI is still mediocre at the hard stuff. That moves the whole risk conversation off capability, where it usually sits, and onto margins and selection. A clean collapse is rare in the wild for a reason. We already built guardrails against exactly these forces: risk adjustment, network-adequacy rules, the limits on physician-owned hospitals, EMTALA, and graduate medical education funding. The model's real use is to say which of those is quietly doing the work, and what happens as it weakens. That the guardrails exist at all is the evidence the force is real, since no one fences off a danger that was never there. Read this way, the unraveling is not a prophecy of bankruptcy but a pressure these institutions currently hold back.

One clarification, because our setting is not the textbook one. The clean version of the story has a single incumbent who once held both the cream and the tail and an entrant who pulls the cream away, so the pool that unravels is that incumbent's own panel. But in the market we have in mind, single-condition telehealth, there often was no such incumbent: the cream and the tail were split across different providers from the start, and the complex patients were never the entrant's to lose. The model does not need the within-provider story. Read Π_I as the books of whoever is left holding the flat-fee patients once the cream is gone, whether that is one safety-net incumbent or the broader primary-care system. In that system-level reading the collapse is not one firm's bankruptcy but rising acuity and cost across the residual sector, with the complex patients unserved or pushed into more expensive care, and the rescue term $\delta(-\Pi_I)$ is exactly the extra public and cross-subsidy financing that keeping them served then takes. The two readings share the same arithmetic; only the unit changes. What both need is a flat-fee cross-subsidy to skim in the first place, which is why the real boundary is the payment regime, not how concentrated the providers are: dentistry is highly fragmented yet largely untrapped because it is paid out of pocket, with little cross-subsidy to pull, while primary and chronic care, paid by a compressed fee, carry a large one.

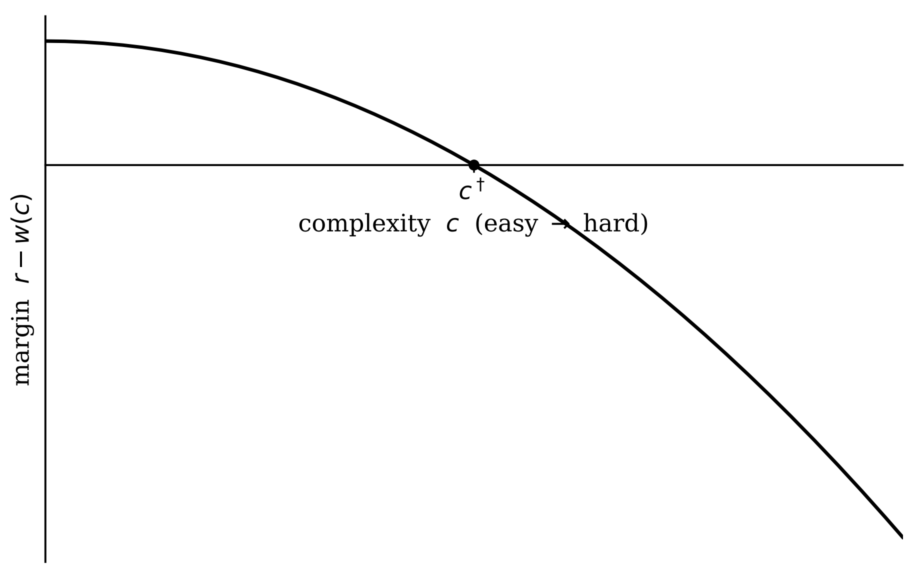

Figure 1: The cost-curve picture (supply-side Einav–Finkelstein). The margin $r - w(c)$ is positive on the routine cream ($c < c^\dagger$) and negative on the tail. Illustrative $w(c) = 0.5 + 2c^2$, $r = 1$, $f = \text{Beta}(2, 4)$ (a shape with most patients easy, tapering to few hard ones), $c^\dagger = 0.5$; AI cost $k(c)$ does not enter this panel.

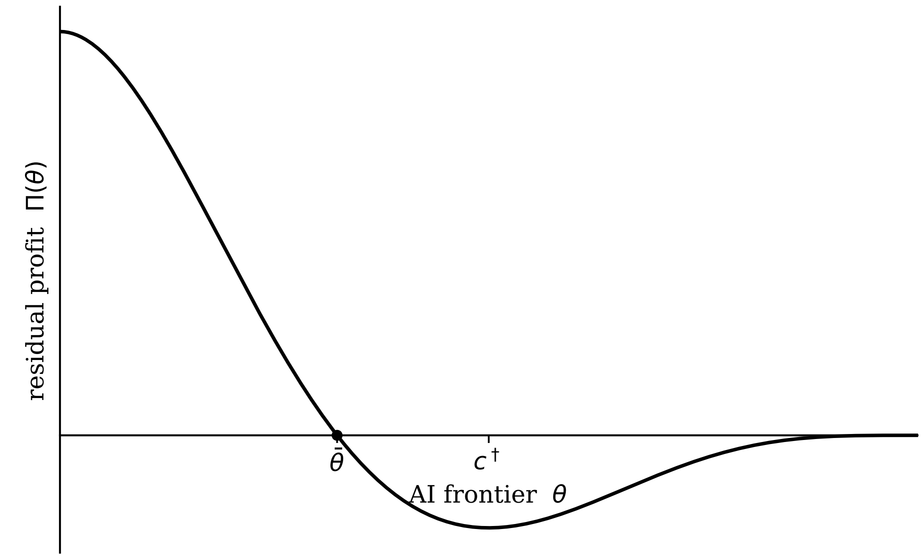

Figure 2: Residual profit $\Pi_I(\theta)$ crosses zero at the unraveling threshold $\bar{\theta} \approx 0.32$, strictly before $c^\dagger = 0.50$ (Proposition 1).

Whether the cream is skimmed, and how deep, depends on *who* does the automating. An **integrated provider** serves the whole panel and automates its own easy cases. A **standalone entrant** serves only an easy band $[0, a]$, $a \le \theta$, at AI cost $k(c)$, and turns the rest away, earning $G(a) = \int_0^a (r - k(c)) f \, dc$. The fee r is just a transfer, so the real resource saving from automating a case is only $w(c) - k(c)$, not $r - k(c)$. The entrant pockets the extra $r - w(c)$, the incumbent's lost margin, which is business stealing (Mankiw and Whinston, 1986).

The social cost of skimming is neither that transfer nor a constant. While the residual pool is still solvent ($a \le \bar{\theta}$), pulling cream out is purely redistributive, and where AI is genuinely cheaper it is also efficient, so society loses nothing (Corollary 1). A real cost switches on only once skimming drives the residual pool underwater ($a > \bar{\theta}$), where covering the abandoned tail takes public funds at the deadweight rate δ (the deficit $-\Pi_I$ is the selection wedge of Einav–Finkelstein and Kong–Layton–Shepard). So the planner maximizes the real saving net of that *state-contingent* rescue cost,

$$\max_a \int_0^a \big(w(c) - k(c)\big) f(c) \, dc \; - \; \delta \max\{0, -\Pi_I(a)\}, \tag{5}$$

while the entrant maximizes $G(a)$ alone, so their stopping points are

$$\underbrace{r - k(a^E) = 0}_{\text{entrant}}, \qquad \underbrace{w(a^*) - k(a^*) = \delta\big(r - w(a^*)\big)\, \mathbf{1}\{a^* > \bar{\theta}\}}_{\text{planner}}. \tag{6}$$

The entrant over-skims, $a^E > a^*$: it automates deeper than is efficient, because it only checks whether AI beats the *fee*, never whether AI beats a *human*, nor whether the pool it leaves behind can still pay its way. We focus on the realistic case where AI is cheaper than a human on the routine cases but grows more expensive than a human on the harder ones, while still undercutting the fee ($k(\bar{\theta}) < w(\bar{\theta})$ and $k(c^\dagger) > r$). Here a planner would keep automating past the break point on cost grounds alone, and stops only because each extra case skimmed deepens the bill to rescue the leftover pool. So the entrant's overshoot is one of *funding*: the band of cases (a^*, a^E) it pulls past $\bar{\theta}$ whose rescue cost it dumps on others. (In the other case, where AI turns expensive before the break point,

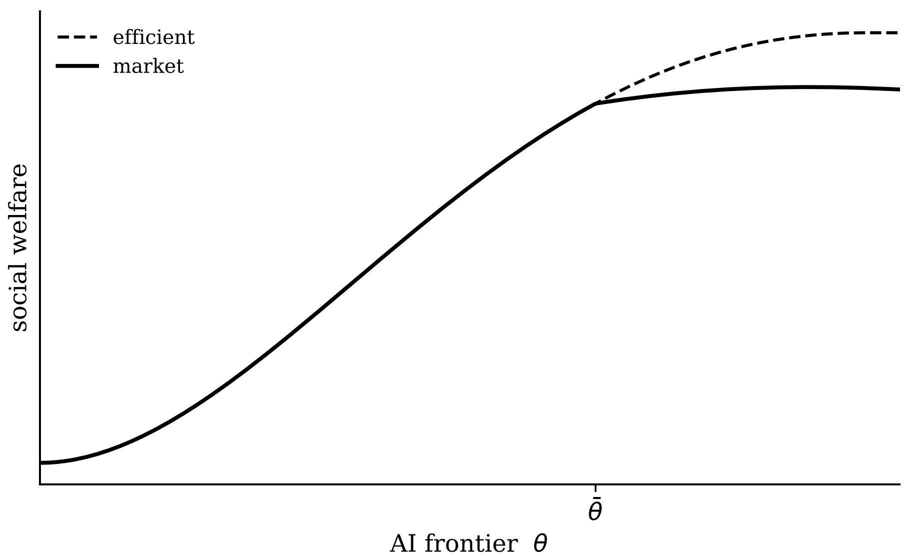

Figure 3: Welfare tracks the first best until $\bar{\theta}$ (skimming is purely redistributive there), then falls below it as the rescue deficit grows continuously from zero, a kink, not a cliff, absent exit frictions (Corollary 1). Shown over the cream $[0, c^{\dagger}]$, with illustrative rising AI cost $k(c)$, $k(0) < w(0)$.

the entrant also overshoots in *depth*, running AI on cases a human would do more cheaply; that one is corrected by a smaller tax, $r - w$, and the funding overshoot is the bigger problem.)

Proposition 2 (The entrant over-skims; integration is the fix)**.** *The entrant skims to the AI break-even frontier $k(a^E) = r$; the planner stops short of it. The corrective per-case tax, charged only where skimming creates real harm, is*

$$\tau(c) = (1 + \delta)\big(r - w(c)\big) \quad \textit{on cases skimmed past } \bar{\theta}, \tag{7}$$

the business-stealing margin $(r - w)$ it captures plus the rescue deadweight $\delta(r - w)$ it externalizes. On the genuinely cheaper, solvent cream the social harm is a pure transfer and the optimal tax is zero, so the levy targets the rescue deficit, not the transfer. What removes the wedge with no tax at all is vertical integration*: an integrated provider automates its own panel, captures $w - k$ directly, steals no margin from itself, and by* (2) *stays solvent, so no rescue deficit ever arises and it automates the efficient amount unprompted.*

In plain terms, the entrant automates deeper than is good for anyone, and there are two ways to stop it: tax the skimming, or let a single organization hold the easy and the hard cases together. So *integration is efficient and fragmentation harmful*: only a provider that holds the cream together with the tail internalizes the cross-subsidy; an entrant that holds only the cream never will. One caution: this says an integrated provider *would* automate efficiently, not that integration restores efficiency in *equilibrium*. By Faulhaber (1975) free entry breaks the cross-subsidy whatever the incumbent's structure, so a standalone entrant can skim an integrated system too. The structural cure is therefore integration *plus* a barrier to cream-only entry (a network-adequacy or must-serve rule), or, equivalently, the skim tax; integration is not self-enforcing against entry. Integrated, salaried, often capitated systems (Kaiser, the VA, academic centers such as UCSF) integrate by design; fragmented small-practice segments do not. Statically the burden falls on complex patients, incumbents, and taxpayers, not the entrant; whether the automators are themselves harmed is dynamic, and the workforce channel answers it.

4 The workforce externality

The financing problem above can, in principle, be fixed with the stroke of a pen, by changing how patients are priced. The problem in this section cannot, because it is about people, and people take a decade to make. If AI does the easy cases, the young clinician never does them, and so never becomes the senior one. We now make that precise, leading with the robust market failure and then the sharper amplifier on top of it.

Write H_t for the stock of senior doctors, the ones who can handle the hard cases and supervise the AI. They are made slowly (about a decade) from routine cases R_t worked through teaching settings, and lost steadily to retirement at rate γ:

$$H_{t+1} - H_t \;=\; g(R_t) - \gamma H_t, \qquad g' > 0,\ g(0) = 0,\ \gamma > 0. \tag{8}$$

In words, the senior stock grows as new seniors are trained from routine cases, $g(R_t)$, and shrinks as old ones retire, γH_t. (Teaching also takes senior time, so a thin stock teaches few; the supervision loop below builds on that.) What is scarce here is not medical-school seats but the cases to learn on: AI does not take the bottom of the ladder, school, it takes the rung just above it. So as routine cases move to AI-native settings that cannot teach, R_t falls toward zero, $g(R_t)$ falls with it, and the senior stock decays. This is the "broken job ladder" (Zhang, 2026) and the "never-skilling" that follows (Natali et al., 2025). The human capacity society still needs, meanwhile, has two parts:

$$D(\theta_t) \;=\; \underbrace{\big(1 - F(\theta_t)\big)}_{\text{complex tail}} \;+\; \underbrace{\omega\, F(\theta_t)}_{\text{AI oversight}}\;, \tag{9}$$

the complex tail AI cannot touch, $1 - F(\theta_t)$, plus a slice ω of senior time to oversee each case AI does handle. This need never falls to zero. Even as AI improves toward doing everything, the oversight slice stays; and if AI plateaus short of the hardest cases, that tail is there for good. Some senior capacity is needed forever, which is Bainbridge's point: automation hands the human exactly the hardest, rarest tasks.

Usually this is not a problem. When a machine takes over a task, the workers it displaces move into new tasks the machine creates, and employment holds up (Acemoglu and Restrepo, 2019). Radiology looks like that so far: the number of radiologists has kept growing, even though task-by-task forecasts predicted cuts (Langlotz, 2025). Here the escape can jam, for two reasons at once. The new job automation creates is supervising the AI, and by (9) that needs *fewer* people, and more senior ones. And the same automation that creates those supervisor jobs drives the routine volume R_t toward zero in (8), drying up the very cases that would have trained the supervisors. The ladder loses a rung at the same moment it asks for more senior people to climb it.

Start with the part that needs no uncertainty at all: left to itself, the market trains no one, and the reason is about who keeps the payoff. Training is expensive, and a trained doctor can work anywhere. So the clinic that pays to train a young doctor does not get to keep the return: the newly minted senior can walk across the street to a competitor, and the value leaks out to every clinic that might hire them. No clinic wants to be the one that foots the bill for everyone else's hires. Economists call this an appropriability problem, and it usually has a fix, the one Becker (1964) described: let the trainee pay for their own training, by taking low wages while they learn in exchange for high wages once trained. Residency runs on exactly this deal. But the deal only works if the high wages later actually arrive, and here they do not, because prices are fixed. When senior doctors are scarce they would normally command a premium, but the flat fee caps what anyone can be paid, so the future reward is too small to repay the training. The trainee cannot finance it and the clinic cannot either, so each one trains as little as it can and the market drives training to zero

($R^{\text{eq}} = 0$, the market half of Proposition 4). And once the routine cases have moved to AI-native settings that cannot teach, even a clinic that wanted to train could not: the cases to learn on are gone.

Two honest caveats. First, we have shut off one escape by assumption. In the bad state, scarce seniors really would get bid up through locum rates, retention bonuses, and productivity pay, none of which the flat fee controls, and that would pull some training back. So the collapse is a *medium-run* result: it binds for as long as that pay response is slower than retirement, which over a decade-long training lag is a long time. The short-run irreversibility of Proposition 3 holds either way. Second, the government already funds training through residency, which is consistent with the market under-providing it, but is not proof, since that funding also reflects politics. Either way, the people problem and the money problem trace to the same root, a fee that does not keep up with complexity, which is the deeper sense in which the two halves are one.

The market failure just described is robust but blunt: the market trains zero. The sharper point is about the right *target*. Even a planner who internalized the whole common pool would still train *more* than enough to cover expected retirements, because AI's ceiling is uncertain and lost expertise cannot be rebuilt in a hurry. The decision is essentially one of insurance: you pay to train now, and the payoff comes only in the bad case, when AI stalls and you need the doctors you can no longer make quickly. To see why, line up three dates. At date 0 someone decides how much training $R \geq 0$ to fund, at cost $\chi(R)$ ($\chi' > 0, \chi'' \geq 0$); that sets the senior stock at date 1, $H_1(R) = (1-\gamma)H_0 + g(R)$. At date 1 we learn how far AI is actually allowed to go, which capability and regulation settle together: with chance p the rosy case, where AI handles almost everything and little human capacity is needed ($D(1) = \omega$), and with chance $1-p$ the bad case, where AI stalls or is held back on the hard cases and much more is needed ($D(\theta_L) > \omega$). The catch is that capacity cannot be rebuilt before date 2, so a shortfall at date 1 is locked in, at a cost of κ per unit short. The planner picks R to minimize expected total cost,

$$\min_{R \geq 0} \; \Phi(R) \;=\; \chi(R) \;+\; \mathbb{E}\big[\kappa\,(D(\theta_\infty) - H_1(R))^+\big], \qquad H_1(R) = (1-\gamma)H_0 + g(R). \tag{10}$$

Here θ_∞ is the realized long-run (plateau) AI frontier, the random variable settled at date 1 that equals 1 in the rosy case (probability p) and θ_L in the bad case (probability $1-p$); the expectation in (10) is taken over it. With the high state slack at $R = 0$ and the low state binding for small R, the interior first-order condition is

$$\chi'(R) \;-\; (1-p)\,\kappa\,g'(R) \;=\; 0 \quad\Longrightarrow\quad R^* > 0 \;\text{ if }\; \chi'(0) < (1-p)\kappa g'(0). \tag{11}$$

The term $(1-p)\kappa g'(R)$ is the **quasi-option value** of keeping training going. It exists only because the bad case can happen (chance $1-p$) and the loss cannot be undone: under certainty ($p = 1$) it vanishes and the planner trains nothing extra. If keeping that option alive is worth enough, the planner trains all the way up to the point where the stock just covers the bad-state need, $R^* = \min\{\text{the interior solution, the gap-closing } R\}$. It helps to separate the two ways AI enters here. That it is already good at the easy cases is certain, and that is what skims the cream and starves training today; whether it ever *gets to do* the hard cases is the uncertain part, the bet p. And that bet is about regulation as much as raw capability: a model good enough for a hard case still cannot be used there if liability rules or regulators do not allow it at that difficulty, as when a government pulls a frontier model from release. The danger is exactly this asymmetry: AI is sure enough on the easy work to gut training now, yet may be kept off the hard work later, whether by its own limits or by rule, and we learn which only once it is too late to rebuild.

Proposition 3 (Irreversibility)**.** *If competition drives $R_t \to 0$ so H_t decays before θ_t plateaus, and the realized long-run need $D(\theta_\infty) > 0$ (which holds whenever $\theta_\infty < 1$ or $\omega > 0$) exceeds the decayed stock, then society needs $H \geq D(\theta_\infty)$ but cannot restore it within the decade-long lag: the shortfall is irreversible on the relevant horizon, with probability at least $1 - p$.*

Put concretely: if competition lets training dry up and AI then fails to master the hard cases, the senior doctors we stopped making cannot be conjured back inside the decade it takes to train one. The shortage is locked in.

Proposition 4 (Option value and under-provision)**.** *The planner's solution to* (10) *sets $R^* > 0$ via* (11)*, while the competitive first-order condition omits the option-value term $(1-p)\kappa g'$ and so sets $R^{\text{eq}} = 0$. The resulting shortfall in training-capable throughput is the full planner level R^*.*

In plain terms: a planner would train extra doctors as insurance against AI stalling, but no single clinic will pay for insurance whose benefit it cannot keep, so the market provides none of it.
With a full distribution of the plateau θ_∞ in place of the two-point one, the option value is $\kappa\, \mathbb{E}[(D(\theta_\infty) - H)^+]$ and R^* rises with a mean-preserving spread of the *need* $D(\theta_\infty)$ (Arrow–Fisher): the more uncertain AI's ceiling, the more reserve capacity is worth holding.

These two failures feed back on each other once we notice that AI has to be *supervised*. Senior time gets pulled three ways: treating the complex tail, supervising the AI (a slice ω of senior time per automated case), and teaching the next generation. The first two are billable and come first, so teaching gets whatever is left. This has a sharp consequence: the amount of AI a system can actually run is capped by how many seniors it has. If supervision alone would use up the senior stock, the usable frontier falls below the technical one, $\theta_{\text{eff}} = F^{-1}(H/\omega) < \theta$. So an AI-native entrant that skims the routine cases is draining the very pool of seniors its own AI needs to supervise it, along with the referral capacity for the cases it cannot handle. That answers the question of who is harmed in the end: over the medium run the externality closes back on the automators, who can no longer run the AI they adopted.

Proposition 5 (Tipping, and what the mandate does)**.** *Left to the market, routine volume migrates to AI-native settings that cannot teach, so the throughput trainees learn on vanishes and new seniors stop being made ($g \to 0$): the only steady state is the trap, $H \to 0$. A training mandate that keeps enough routine volume in teaching-capable settings restores $g > 0$, and a* healthy *steady state then exists provided training can outrun retirement,* $\max_H \left[g(H - D(\theta)) - \gamma H\right] \geq 0$. *When that holds, the healthy state is separated from the trap by an unstable threshold, so the mandate does not merely nudge the system toward the good equilibrium, it is what makes the good equilibrium* exist *at all; when it fails (retirement outpaces the best attainable training yield), no mandate suffices and the only remedy is to lower the need D itself, for example a smaller oversight load ω. The case for the mandate does not rest on the bad state being likely: once routine volume is gone, the collapse is deterministic over the medium run.*

Plainly: without a rule keeping teaching cases in the system, training falls to zero and the trap is the only outcome; a training mandate is what makes a healthy outcome possible at all, so long as training can keep up with retirements.

Remark (the mechanics behind the loop). The cap works through how senior time is rationed. Running AI at the frontier θ uses up $D(\theta) = (1 - F(\theta)) + \omega F(\theta)$ of senior time, treating the tail plus supervising, and because more automation frees seniors from routine work this need *falls* as θ rises. Three cases follow. If the senior stock H covers $D(\theta)$ with room to spare, the surplus $H - D(\theta)$ goes to teaching. If it covers supervision but not the whole tail, the AI still runs at full frontier

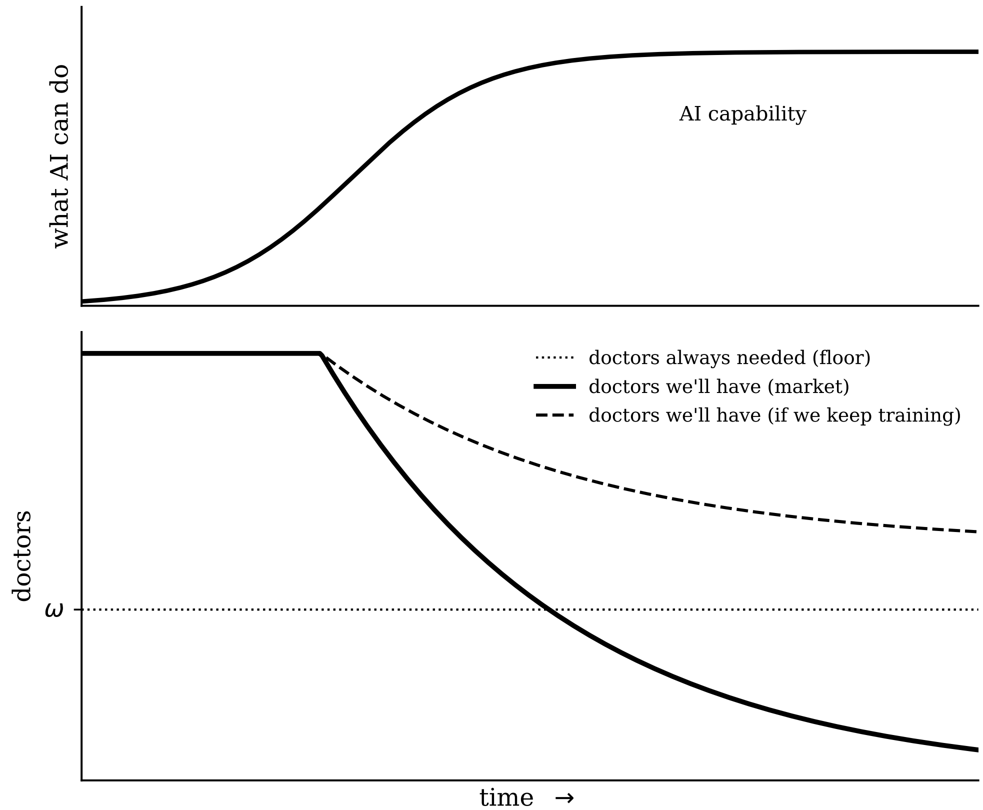

Figure 4: **Fast machine, slow people.** *Top:* AI capability rises fast, then plateaus (the driver). *Bottom:* the senior capacity AI cannot replace is a permanent floor ω; left to the market (solid) training is starved and supply decays *below* the floor, a shortage that cannot be rebuilt in time, while keeping training up (dashed) holds it above. The shapes are the capacity law $\dot{H} = g(R) - \gamma H$: with training starved ($R \to 0$) supply decays exponentially at the retirement rate γ; with training held at R^* it approaches $g(R^*)/\gamma$. Levels are illustrative (Propositions 3–4).

but some hard cases go unserved and nothing is left to teach, the visible warning sign before the trap. And if the stock cannot even cover supervision, the usable frontier itself is forced down to θ_{eff}. New seniors need *both* a teacher's time and routine cases to learn on, so training runs at the pace of whichever is scarcer, $g = g(\min\{R_{\text{retained}}, H_{\text{teach}}\})$ with $H_{\text{teach}} = \max\{H - D(\theta), 0\}$; once the routine cases are gone, teachers alone produce no one. In steady state, new seniors just balance retirements, $g(\cdot) = \gamma H$.

5 The coupling and policy

Sections 3 and 4 share one object: the routine cases are the cream that finances complex care *and* the throughput that trains specialists. Automating them under competition both withdraws the cross-subsidy and starves the pipeline. Policy needs two tools, not one, because a single structural cure reaches only part of the problem.

Proposition 6 (The coupling, and why transfers and administered prices alone cannot do both)**.** *Both failures grow from the same primitive, flat pricing, but they do not yield to the same lever.*

Vertical integration is necessary and sufficient for the financing externality*: an integrated provider captures $w - k$, withdraws no cross-subsidy, and stays solvent (Proposition 2), and because it needs $D(\theta)$ seniors for its own tail and oversight it also trains enough to replace its own retirements ($g(R) = \gamma H$ at $H = D(\theta)$; Kaiser runs residencies).* It is necessary but not sufficient for the workforce externality*: what no market structure prices is the option value* $(1-p)\kappa g'$ *of training* above *replacement against a bad-state plateau, because in the bad state that capacity spills out through referral markets and labor mobility, a public good even for a single integrated system. So the two externalities do not pull in opposite directions on market structure; rather, structure settles one of them and only part of the other. Within the class of instruments we analyze, no transfer or administered price reaches both margins, since the financing margin depends on (r, w, k) and the option-value margin on the disjoint set (p, κ, g, γ); so two instruments suffice (Tinbergen, 1952).*

We do not claim that *no* single instrument could ever address both margins. A more elaborate instrument outside the transfer/administered-price class we model, for instance a complexity-indexed fee whose proceeds are earmarked to fund training slots, could in principle touch both the financing margin (through how it prices complexity) and the option-value margin (through how its earmark funds reserve training), and is not ruled out by the model. Whether one such well-designed instrument can implement the planner's allocation is left to future work; our point is the narrower one that the two failures are distinct and that the simple levers, pure transfers and flat administered repricing, do not reach both.
In plain terms: the two failures come from one cause but need two different cures. Making providers bigger fixes the money problem and covers the ordinary training a system needs to replace its own retirements, but it cannot buy the *insurance* training against a bad AI outcome, so a separate training rule is still needed. This is the precise content of what we loosely call the "antitrust reversal," and the label should be read with care. The integration result is nothing more exotic than the standard cross-subsidy internalization of Faulhaber (1975): a provider that holds the cream together with the tail bears the full cost of pulling cream out and so does not over-skim. It is not a general endorsement of concentration, and it is not self-enforcing, since by Faulhaber free entry can break the cross-subsidy whatever the incumbent's structure, so integration still needs a barrier to cream-only entry or the skim tax to hold up against entry (consistent with the caveat in Section 3). Integrating the cream with the tail removes the financing externality, the reverse of the usual worry that concentration is harmful, but it is *not* a claim that concentration is good for the workforce: integration serves the people problem only up to replacement, and where prices are negotiated rather than administered it can make matters worse (Section 6). Two tools follow, one for each failure (Table 1).

Start with the money problem. It exists only because the fee is flat while costs rise, so the clean fix is to pay more for sicker patients. Set $r(c) = w(c) + m$, a fee that tracks cost plus a margin, and the incumbent's leftover profit becomes $\Pi_I(\theta) = m(1 - F(\theta)) \geq 0$ for every θ: there is no cream to skim, the pool never goes broke, and the problem disappears. This is the optimal-risk-adjustment result of Glazer and McGuire (2000), now under automation pressure. The catch is that sickness can never be measured perfectly. If the payer can price only a fraction ϕ of the true cost difference, a residual skim incentive survives, $\sigma(\phi) = \delta(1-\phi)\Delta$ (where Δ is the unpriced cream per routine case), fading to zero only as measurement becomes perfect ($\phi \to 1$). That is what the evidence shows: risk adjustment reshapes selection rather than removing it (Brown et al. 2014; Cabral, Geruso and Mahoney 2018), and the ceiling on ϕ is upcoding (Geruso and Layton, 2020; Shepard, 2022). So measuring better and taxing the residual are two dials on one problem: code as finely as you can, then put the skim tax $\tau(c) = (1+\delta)(r - w(c))$ of Proposition 2 on what is left. One honest limit. The ideal per-case tax needs the regulator to know each case's true cost $w(c)$, the very thing it cannot measure, so in practice the tax is second-best, levied on the entrant's *observable* panel mix, how skewed toward easy cases its book is. It inherits a milder form of the same gaming problem: an entrant can take a thin slice of complex cases to dilute its measured mix while quietly referring the hard work back out. The levy is only ever as good as the mix the payer can verify.

No pricing fix reaches the people problem, because that one is about the stock of senior doctors, not this period's prices. The second tool is a mandate to keep training going, $R_t \geq R^*$, which restores the planner's training condition (11) and the option value with it. This is the move aviation made with autopilot-dependent pilots, who are now required to hand-fly a stretch periodically. And it is not hypothetical: Medicare's graduate medical education program already pays for exactly this throughput, over \$16 billion a year (Congressional Research Service, 2022), a subsidy whose rationale our model supplies and whose design, tying funded slots to routine volume in teaching-capable settings, it would sharpen. Redistribution, by contrast, does nothing here. A pure lump-sum transfer leaves all three of the things that matter untouched, the collapse point $\bar{\theta}$, the skimming depth a^E, and equilibrium training R^{eq}, because none of them depends on it. So universal basic income, retraining, capital-income taxes, and worker equity move money around but fix neither externality. (A tax that distorted the AI cost k would move a^E; the point is about pure transfers, not every conceivable tax.)

Table 1: Two failures, two instruments (Tinbergen mapping).

Failure	Nature	Corrective instrument
Cross-subsidy withdrawal (selection)	Static	Risk-adjusted reimbursement (first best); Pigouvian skim tax (second best)
Training-throughput erosion	Dynamic, irreversible	Training-capacity / GME mandate (aviation analog)
Integration vs. fragmentation	Structural	Fixes financing and replacement training; not the option value
Displacement of clinicians	Distributional	UBI / retraining / capital tax: none bind

How early does the collapse come? Early, and for a reason that has little to do with how clever the AI is. Appendix B works this out from public data: take the common office-visit codes 99211 through 99215 as five rungs of complexity, use each code's typical visit time as a stand-in for its cost, and treat the fee as flat across them. The tipping point then turns almost entirely on one number, the operating margin a primary-care practice runs on. At a comfortable ten-percent margin the pool can lose roughly its easiest two-fifths of visits before it tips; at the low-single-digit margins primary care actually runs (MGMA and AAFP surveys), it tips after well under a fifth of routine volume is skimmed. These shares are illustrative, not an empirical estimate; the qualitative pattern, not the specific fraction, is the result. The result holds even when payment partly tracks complexity, and disappears only when payment tracks it almost perfectly, which is the sign that the live margin is the flatness *within* a payment cell, which is exactly what Assumption 1 assumes. The binding input is the operating margin, not AI capability.

6 Discussion

Every modeling choice we made tends to make the problem look smaller, so our result is a floor, not a worst case. We model one sector, but the complex patients turned away do not disappear; they show up later, sicker, in costlier settings. We treat the limit of what AI can safely do as fixed, but if entrants push AI past that limit and pass the mistakes to others, the over-automation is worse than we say. And the whole argument rests on one assumption, that payment is roughly flat across complexity within a billing category. Where payment already tracks complexity well the problem shrinks, but paying for complexity is exactly the fix we recommend.

We are equally clear about what we do not claim. Our accounting adds up costs, not the convenience patients get from faster, cheaper, easier care through AI, and that gap cuts in our favor: if patients value the convenience enough, some skimming is genuinely worth it and the case for stepping in is weaker than the cost numbers alone suggest. But we leave costs out too. Patients turned away resurface later in more expensive care, entrants offload the complex patients they took on by mistake, and AI used past what it can safely handle does real harm, all of which pushes the other way. So we do not pretend to know the net effect. What we show is the mechanism and the cost side of it; a full verdict would need a richer account of what patients want, which we have deliberately kept to a single assumption.

Four caveats sharpen the result rather than weaken it. First, the story needs human oversight of AI to stay necessary; we assume it is required case by case, which is how clinical-AI rules and liability work today, and if oversight instead became almost free the supervision half of the trap would fade, leaving the weaker training problem. Second, real skimming runs along insurance type, not just case difficulty: providers already favor better-paying patients (Werbeck, Wübker and Ziebarth, 2021), and AI-native entrants will do both at once, which only makes the collapse arrive sooner. Third, real skimming is messier than our clean cut, so entrants take on complex patients by mistake and offload them mid-treatment, which is worse than the tidy version we model. Fourth, our structural fix, letting one organization hold the easy and hard cases together, works only where the payer sets prices; where prices are negotiated, large systems tend to raise them without improving care (Gaynor and Town, 2011), so there the fix should run through paying for complexity and taxing the skim instead.

If we are right, the visible pattern is not the mass layoff of clinicians but a reshuffle. AI first assists, then becomes the front door for routine care, while clinicians move toward the complex, hands-on, relational, and oversight work that is left. The part unique to our story is on the training side: as routine cases drain out of teaching settings, fewer juniors get the repetitions that make a specialist, and skill erodes most where AI is used most (Natali et al., 2025; Budzyń et al., 2025). The rising cost and acuity of the patients left behind, like the likely jump in total visits once each one is cheaper (Pauly, 1968), is something other models predict too, so neither alone pins down our mechanism. Everything rests on whether AI can actually make the routine clinical call, and that is the question our companion study of week-8 GLP-1 follow-up visits is built to answer. It is where the theory meets data.

None of the pieces here are new on their own: skimming and risk adjustment are old ideas, and the erosion of training by automation is its own active field. What is new is the object that ties them, one set of routine cases that funds hard care and trains the people who provide it, so automating it breaks both at once and needs two separate fixes.

The plain version is this. Automating routine clinical work is a good thing. A clinic that adopts AI and keeps its patients comes out ahead and can fund its complex care more easily than before. The trouble is not the automation; it is who does it and what they leave behind. A newcomer that automates only the easy patients and turns away the hard ones takes the cream and leaves the bill, and no clinic competing for those same easy patients can afford to be the one that holds the whole panel together. Meanwhile the ordinary cases that taught every clinician their judgment are gone, and a senior doctor cannot be conjured on short notice. That is the trap: each step makes sense on its own, the result does not, and part of it cannot be undone. The answer is not to slow the technology down. It is to pay for complexity, so there is no cream to skim, and to require those who automate the routine to keep alive the human capacity that automation quietly spends.

A Proofs

Maintained: Assumption 1; f continuous and positive; w continuous, strictly increasing; k increasing and convex with $k(0) < w(0)$; $v(c) > w(c)$, $v(c) > k(c)$ for $c \leq \theta$.

Proof of Proposition 1 and Corollary 1. Π_I in (3) is continuous; $\Pi_I(0) = r - \bar{w} > 0$ by Assumption 1. At $c^\dagger$, $\Pi_I(c^\dagger) = \int_{c^\dagger}^1 (r-w)f\,dc < 0$ since $r - w < 0$ on $(c^\dagger, 1]$ over a set of positive measure. By the intermediate value theorem $\exists\,\bar{\theta} \in (0, c^\dagger)$ with $\Pi_I(\bar{\theta}) = 0$; from (3), $\Pi_I'(\theta) = -(r - w(\theta))f(\theta) < 0$ for $\theta < c^\dagger$, so $\bar{\theta}$ is unique there and $\Pi_I < 0$ just above it. Dividing $\Pi_I(\bar{\theta}) = 0$ by $1 - F(\bar{\theta})$ gives $\mathbb{E}[w \mid c > \bar{\theta}] = r$. For the corollary: for $\theta \leq \bar{\theta}$ the allocation equals the planner's and welfare equals the first best, differing only in surplus division; for $\theta > \bar{\theta}$ the tail is unserved (loss $\int_\theta^1 (v-w)f$) or rescued (deadweight $\delta(-\Pi_I(\theta))$), giving (4). Because $-\Pi_I(\bar{\theta}) = 0$, the loss has no jump at $\bar{\theta}$. Its left-derivative is 0 and its right-derivative is $\delta(-\Pi_I'(\bar{\theta})) = \delta(r - w(\bar{\theta}))f(\bar{\theta})$, strictly positive because $w(\bar{\theta}) < r$ ($\bar{\theta} < c^\dagger$ by Proposition 1). The loss is thus continuous but opens *linearly* (first order), not as a second-order triangle, and welfare has a downward kink at $\bar{\theta}$ rather than a cliff; the loss then grows at rate $\delta(r - w(\theta))f(\theta)$. A discontinuity requires a separate friction (exit-driven capital loss or a rescue lag), not the frictionless model. □

Proof of Proposition 2. The entrant bears none of the rescue cost, so from (6) it skims while $r - k(a) > 0$, to the break-even $k(a^E) = r$. For the planner, differentiate (5) using $\frac{d}{da}\max\{0, -\Pi_I(a)\} = (r - w(a))f(a)\,\mathbf{1}\{a > \bar{\theta}\}$ (from (3), since $\Pi_I(a) \gtrless 0$ as $a \lessgtr \bar{\theta}$ for $a < c^\dagger$): the marginal value of skimming case a is $\big(w(a) - k(a)\big)f(a) - \delta\big(r - w(a)\big)f(a)\,\mathbf{1}\{a > \bar{\theta}\}$. In the crossing regime $k(\bar{\theta}) < w(\bar{\theta})$ and $k(c^\dagger) > r$, AI is cheaper than a human through $\bar{\theta}$, so for $a \leq \bar{\theta}$ the marginal value is $w - k > 0$: the planner automates the whole solvent band, where skimming is a pure transfer and the corrective tax is zero. Past $\bar{\theta}$ the planner stops where $w(a^*) - k(a^*) = \delta(r - w(a^*))$, interior because $k(\bar{\theta}) < w(\bar{\theta})$ keeps the left side positive at $\bar{\theta}$ while $k(c^\dagger) > r$ turns it negative before $c^\dagger$ (a unique interior root needs $k' > (1+\delta)w'$ there). The entrant keeps going, since $r - k(a^*) = (w(a^*) - k(a^*)) + (r - w(a^*)) = (1+\delta)(r - w(a^*)) > 0$; hence $a^E > a^*$. A per-case tax shifts the entrant's stopping condition to $r - k(a) - \tau(a) = 0$; evaluating at a^* gives $\tau(a^*) = r - k(a^*) = (1+\delta)(r - w(a^*))$, which is (7). (Outside this regime, if AI turns dearer than a human before $\bar{\theta}$, the binding margin is depth and the implementing tax is the smaller $r - w(a^*)$.) □

Proof of Propositions 3 and 4. Suppose $R^{\text{eq}} = 0$ (established last). Then $H_1 = (1-\gamma)H_0$; if the low state obtains and $H_1 < D(\theta_L)$, the shortfall cannot be eliminated before date 2 by irreversibility, so $\kappa(D(\theta_L) - H_1)$ is incurred and unrecoverable; since $D(\theta_L) > \omega > 0$ the need is positive in the low state, so a positive irreversible shortfall occurs with probability $\geq 1 - p$, proving Proposition 3. For Proposition 4: differentiating (10) where only the low state binds gives (11), so $R^* > 0$ when $\chi'(0) < (1-p)\kappa g'(0)$; the term $(1-p)\kappa g'(R)$ is the quasi-option value (zero under $p = 1$). The firm's payoff omits the shortfall term, which falls on the common pool H and is not privately appropriable, so each firm minimizes $\chi(R)$ and sets $R^{\text{eq}} = 0$; because that term enters the planner's first-order condition but not the firm's, the quantity gap is the full $R^* - R^{\text{eq}} = R^*$. □

Proof of Proposition 6. An integrated provider solving (2) automates the efficient set and, holding the whole panel, withdraws no cross-subsidy, so the financing wedge of Proposition 2 vanishes; needing $D(\theta)$ seniors, it also funds replacement throughput ($g(R) = \gamma H$ evaluated at $H = D(\theta)$). But its private objective still omits the bad-state shortfall in (10), which spills across firms through referral and labor markets, so it provides no training *above* replacement and the option value $(1-p)\kappa g'$ stays unpriced (Proposition 4). Conversely, a throughput floor $R \geq R^*$ supplies that option value but enters neither $\bar{\theta}$ nor the skim tax (7). The financing margin depends on (r, w, k) and the option-value margin on (p, κ, g, γ); the two parameter sets are disjoint, so a single instrument that corrected both would have to enter both, which no transfer or administered price does. By Tinbergen (1952), two instruments are required. □

B Calibration

The headline of Section 5, that the collapse point turns on the operating margin rather than on AI capability, can be taken off the back of an envelope and onto public data. Treat the established-patient office-visit codes 99211 through 99215 as five rungs of complexity, and proxy the cost of each by its typical visit time (midpoints of the 2021 office-visit time bands, roughly 5, 15, 25, 35, and 47 minutes). Weight them by their approximate national shares, (0.02, 0.06, 0.35, 0.45, 0.12), which put most volume on 99213 and 99214. The mean cost is then $\bar{w} \approx 31$ minutes.

Let the flat fee be $r = m\,\bar{w}$, where m is one plus the operating margin. Skimming takes the cheapest codes first; the residual pool collapses when the average cost of the codes that remain first exceeds r, and $F(\bar{\theta})$ is the share of visits skimmed at that point. This is one line of arithmetic per margin, reported in Table 2.

Table 2: Collapse threshold by operating margin (discrete E/M calibration). $F(\bar{\theta})$ is the share of visits skimmed when the residual pool's average cost first exceeds the fee.

Operating margin	2%	5%	10%
$F(\bar{\theta})$	0.08	0.08	0.43

The pattern is the point. At a comfortable ten-percent margin the pool absorbs skimming up through the mid-complexity visit, losing about 0.43 of volume before it tips; at a one-percent margin it tips at 0.02. At the low-single-digit margins primary care actually runs (MGMA and AAFP cost surveys put operating margins in the low single digits), it tips after only about 0.08 of volume, well under a fifth. The threshold moves with the margin, not with how much of medicine AI can do. These figures are illustrative, not empirical estimates. Note also that the discrete five-point grid reports the same 0.08 share at both the 2% and 5% margins: this equality is a grid artifact, since both margins tip on the same rung of the coarse five-code ladder, and the continuous-model check below resolves it into distinct values (0.05 versus 0.13).

Two checks. First, the same exercise on the continuous illustrative model of the figures ($w(c) = 0.5 + 2c^2$, $f = \text{Beta}(2, 4)$) gives a smooth version of the same curve: $F(\bar{\theta}) \approx 0.05$ at a two-percent margin, 0.13 at five percent, and 0.24 at ten percent. Second, the result survives partial complexity-pricing. Let payment tilt part-way toward cost, $r(c) = m[(1-\rho)\bar{w} + \rho\, w(c)]$, with $\rho = 0$ flat and $\rho = 1$ fully cost-proportional. Figure 5 traces $F(\bar{\theta})$ against the margin for several ρ. A partial tilt ($\rho = 0.3$) still leaves an early collapse: at a five-percent margin $F(\bar{\theta})$ rises only from 0.13 to 0.18. Only near-full tilt ($\rho \to 1$) removes it. That is the precise content of Assumption 1: across-code payment is already largely priced, so the margin that matters is the flatness *within* a payment cell.

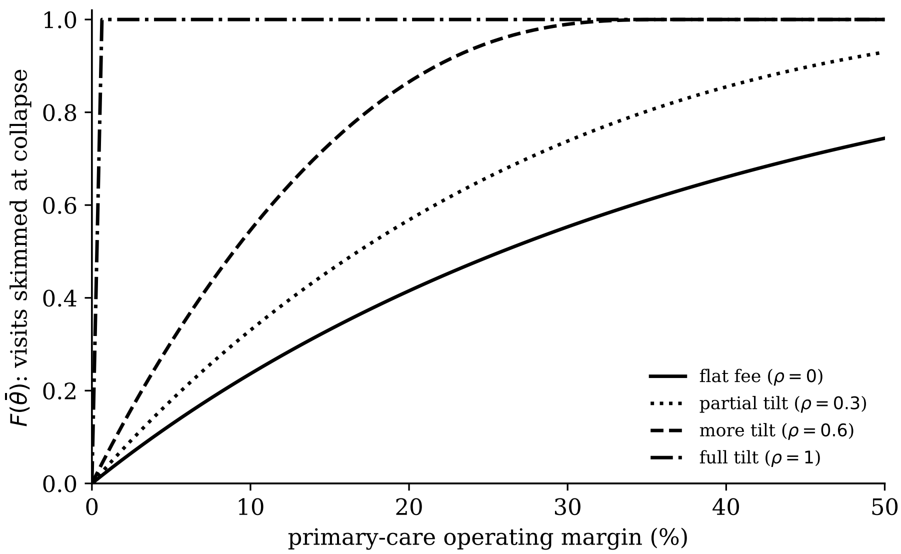

Figure 5: The binding input is the operating margin, not AI capability. $F(\bar{\theta})$, the share of visits skimmed when the residual pool collapses, against the primary-care operating margin, at payment-tilt fractions ρ ($\rho = 0$ flat fee, $\rho = 1$ fully cost-proportional). At the thin margins primary care runs, collapse is early and survives partial tilt; only near-full tilt eliminates it. Continuous illustrative model ($w = 0.5 + 2c^2$, $f = \text{Beta}(2, 4)$).

One caveat keeps the exercise honest. Visit time is a coarse proxy for cost, and the within-cell cost dispersion the mechanism ultimately needs is only partly observable. Time-driven activity-based costing studies and the visit-duration distributions in NAMCS both give within-code variation in cost, and a fuller calibration would use them to pin down $w(c)$ and $f(c)$ inside a single code.

Conflicts of interest / disclosure

The author is a contractor for Mochi Health, a telehealth company operating in the single-condition care market this paper models. Mochi Health had no role in this work, which is theoretical; no external funding was received.

References

Acemoglu, D., & Restrepo, P. (2018). The race between man and machine. *American Economic Review, 108*(6), 1488–1542.

Acemoglu, D., & Restrepo, P. (2019). Automation and new tasks. *Journal of Economic Perspectives, 33*(2), 3–30.

Afrouzi, H., Blanco, A., Drenik, A., & Hurst, E. (2026). *Automation, learning, and career dynamics* (NBER Working Paper 35157).

Akerlof, G. A. (1970). The market for "lemons." *Quarterly Journal of Economics, 84*(3), 488–500.

Armstrong, M. (2001). Access pricing, bypass, and universal service. *American Economic Review, 91*(2), 297–301.

Arrow, K. J. (1963). Uncertainty and the welfare economics of medical care. *American Economic Review, 53*(5), 941–973.

Arrow, K. J., & Fisher, A. C. (1974). Environmental preservation, uncertainty, and irreversibility. *Quarterly Journal of Economics, 88*(2), 312–319.

Autor, D. H. (2015). Why are there still so many jobs? *Journal of Economic Perspectives, 29*(3), 3–30.

Autor, D. H., Levy, F., & Murnane, R. J. (2003). The skill content of recent technological change. *Quarterly Journal of Economics, 118*(4), 1279–1333.

Bainbridge, L. (1983). Ironies of automation. *Automatica, 19*(6), 775–779.

Barro, J. R., Huckman, R. S., & Kessler, D. P. (2006). The effects of cardiac specialty hospitals on the cost and quality of medical care. *Journal of Health Economics, 25*(4), 702–721.

Bayraktar, E. (2026). *The demand externality of automation.* arXiv:2605.05127.

Becker, G. S. (1964). *Human capital.* Columbia University Press (for NBER).

Brown, J., Duggan, M., Kuziemko, I., & Woolston, W. (2014). How does risk selection respond to risk adjustment? *American Economic Review, 104*(10), 3335–3364.

Budzyń, K., Romańczyk, M., Kitala, D., et al. (2025). Endoscopist deskilling risk after exposure to artificial intelligence in colonoscopy: a multicentre, observational study. *Lancet Gastroenterology & Hepatology, 10*(10), 896–903.

Choné, P., Flochel, L., & Perrot, A. (2002). Allocating and funding universal service obligations in a competitive market. *International Journal of Industrial Organization, 20*(9), 1247–1276.

Cabral, M., Geruso, M., & Mahoney, N. (2018). Do larger health insurance subsidies benefit patients or producers? *American Economic Review, 108*(8), 2048–2087.

Congressional Research Service. (2022). *Medicare graduate medical education payments: An overview* (CRS Report IF10960). `https://www.congress.gov/crs-product/IF10960`

Cutler, D. M., & Zeckhauser, R. J. (2000). The anatomy of health insurance. In *Handbook of Health Economics*, vol. 1A.

Dixit, A. K., & Pindyck, R. S. (1994). *Investment under uncertainty.* Princeton University Press.

Dranove, D., & Garthwaite, C. (2022). *Artificial intelligence, the evolution of the healthcare value chain, and the future of the physician* (NBER WP 30607).

Einav, L., & Finkelstein, A. (2011). Selection in insurance markets: Theory and empirics in pictures. *Journal of Economic Perspectives, 25*(1), 115–138.

Einav, L., Finkelstein, A., & Cullen, M. R. (2010). Estimating welfare in insurance markets using variation in prices. *Quarterly Journal of Economics, 125*(3), 877–921.

Faulhaber, G. R. (1975). Cross-subsidization: Pricing in public enterprises. *American Economic Review, 65*(5), 966–977.

Geruso, M., & Layton, T. (2020). Upcoding: Evidence from Medicare on squishy risk adjustment. *Journal of Political Economy, 128*(3), 984–1026.

Gaynor, M., & Town, R. J. (2011). Competition in health care markets. In *Handbook of Health Economics*, vol. 2.

Glazer, J., & McGuire, T. G. (2000). Optimal risk adjustment in markets with adverse selection. *American Economic Review, 90*(4), 1055–1071.

Hemenway Falk, B., & Tsoukalas, G. (2026). *The AI layoff trap.* arXiv:2603.20617.

Henry, C. (1974). Investment decisions under uncertainty: The irreversibility effect. *American Economic Review, 64*(6), 1006–1012.

Kong, E., Layton, T., & Shepard, M. (2024). *Adverse selection and (un)natural monopoly in insurance markets* (NBER WP 33187).

Langlotz, C. P. (2025). *The effect of AI on the radiologist workforce: A task-based analysis.* medRxiv.

Mankiw, N. G., & Whinston, M. D. (1986). Free entry and social inefficiency. *RAND Journal of Economics, 17*(1), 48–58.

Natali, C., Marconi, L., Dias Duran, L. D., Miglioretti, M., & Cabitza, F. (2025). AI-induced deskilling in medicine: A mixed-method review and research agenda for healthcare and beyond. *Artificial Intelligence Review, 58*(11), 356.

Newhouse, J. P. (1996). Reimbursing health plans and health providers. *Journal of Economic Literature, 34*(3), 1236–1263.

Panzar, J. C., & Willig, R. D. (1977). Free entry and the sustainability of natural monopoly. *Bell Journal of Economics, 8*(1), 1–22.

Pauly, M. V. (1968). The economics of moral hazard: Comment. *American Economic Review, 58*(3), 531–537.

Rothschild, M., & Stiglitz, J. (1976). Equilibrium in competitive insurance markets. *Quarterly Journal of Economics, 90*(4), 629–649.

Sahni, N., Stein, G., Zemmel, R., & Cutler, D. M. (2023). *The potential impact of artificial intelligence on healthcare spending* (NBER WP 30857).

Shepard, M. (2022). Hospital network competition and adverse selection. *American Economic Review, 112*(2), 578–615.

Tinbergen, J. (1952). *On the theory of economic policy.* North-Holland.

Valletti, T. M., Hoernig, S., & Barros, P. P. (2002). Universal service and entry. *Journal of Regulatory Economics, 21*(2), 169–190.

van de Ven, W. P. M. M., & Ellis, R. P. (2000). Risk adjustment in competitive health plan markets. In *Handbook of Health Economics*, vol. 1A.

Werbeck, A., Wübker, A., & Ziebarth, N. R. (2021). Cream skimming by health care providers and inequality in health care access: Evidence from a randomized field experiment. *Journal of Economic Behavior & Organization, 188*, 1325–1350.

Zhang, M. B. (2026). *The broken job ladder.* Working paper, USC Marshall School of Business.